

Quotient Laws of Exponents*Question Bank 3.02*

Show your work. Circle your final answer.

For questions 1 through 3, show the quotient of powers law, then evaluate.

1. $\frac{10^5}{10^3}$

2. $\frac{8^4}{8^1}$

3. $\frac{(-6)^6}{(-6)^5}$

4. When dividing two powers that have the same nonzero base, keep the base and _____ the exponents.

For questions 5 and 6, apply the power of a quotient law. Do not simplify the fraction.

5. $\left(\frac{-5}{10}\right)^7$

6. $\left(\frac{3}{-4}\right)^5$

The zero exponent.

7. Fill in the missing value in the pattern: $4^4 = 256$, $4^3 = 64$, $4^2 = 16$, $4^1 = 4$,
 $4^0 =$ _____

8. Evaluate 9^0 and explain in one sentence why your answer is not 0.

For questions 9 through 11, rewrite the expression using a law of exponents, name the law, then evaluate.

9. $\frac{7.6^5}{7.6^3}$

10. $\left(\frac{10}{11}\right)^0$

11. $\left(\frac{0.8}{1.6}\right)^4$

Working backwards.

12. Find the value of n that makes each statement true.

a. $\frac{3^7}{3^n} = 3^2$

b. $\left(\frac{5}{2}\right)^n = \frac{125}{8}$

c. $\frac{6^n}{6^4} = 1$

13. Which expression is equivalent to $\frac{12^9}{12^3}$?

- ☐ A. 1^6
- ☐ B. 12^6
- ☐ C. 12^3
- ☐ D. 12^{12}

14. Select ALL of the expressions whose value is 1.

☐ A. $3^0 \cdot 3$

☐ B. 5^0

☐ C. 1^5

☐ D. $\frac{3^4}{3^4}$

☐ E. $(-4)^0$

☐ F. $\frac{3^4}{3^3}$